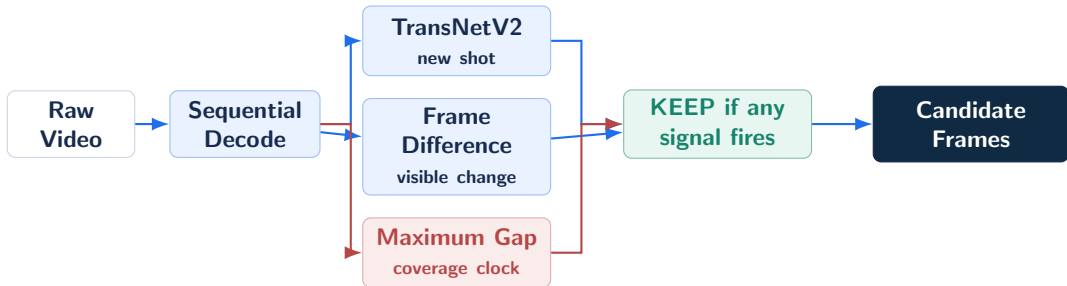


AIC 2026 Frame Preprocessing



Preprocessing ends at a compact, traceable FrameStore.

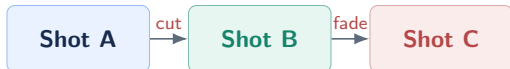
Offline Selection — What Runs on Every Frame



$$\text{candidate}_t = \text{cut}_t \vee \text{diff}_t > \tau \vee (t - t_{\text{last}}) \geq G$$

Evidence basis: [TransNetV2](#) proposes shot boundaries; [MGSampler](#) shows motion-guided sampling can beat fixed sampling.

Paper 1 — Why TransNetV2?



Paper overview: TransNetV2 detects where one video shot changes into another, including both cuts and gradual transitions.

Contribution: one pretrained model for hard cuts and gradual transitions; synthetic transitions strengthen training

Paper scope: shot boundaries, not actions inside a shot.

Published shot-boundary F1

Dataset	F1
ClipShots	77.9
BBC Planet Earth	96.2
RAI	93.9

Evidence: strong cross-dataset F1 for shot detection.

Our decision: use it only to propose boundaries; keep a small neighborhood and tune the threshold on AIC.

Conclusion: TransNetV2 gives a reliable low-resolution scene-change signal, so new shots are preserved before later deduplication can remove similar frames.

Souček & Lokoč, TransNet V2, 2020.

Paper 2 — Motion-Guided Sampling

MGSampler insight



Paper conclusion: fixed-rate sampling wastes frames in static regions and undersamples motion-heavy regions.

Simple idea: keep fewer frames in static parts and more frames where visible actions are happening.

Our lightweight adoption

$$d_t = \text{mean}(|l_t - l_{t-1}|)$$

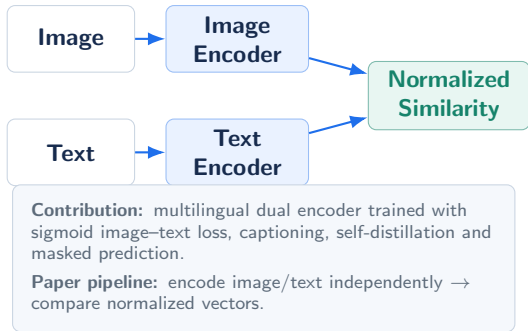
Use Frame Difference: cheap, streaming, and requires no additional model or training.

KFS-Bench: precision alone is insufficient; scene coverage and sampling balance also affect QA.

Add Maximum Gap: force periodic candidates when motion is weak.

Decision: KEEP a local motion peak OR when Maximum Gap is reached.
MGSampler, 2021 — KFS-Bench, WACV 2026

Paper 3 — Why SigLIP2?



Published Recall@1 — B/16 @ 224

Benchmark	SigLIP	SigLIP2
COCO T→I	47.2	52.1
XM3600 T→I	22.4	40.3
XM3600 I→T	29.3	50.7

Decision: use one SigLIP2 image vector for local semantic dedup and preserve it for later retrieval.

Role: convert each candidate frame into a semantic vector. Similar vectors indicate similar visual content, enabling semantic deduplication.

Tschannen et al., SigLIP 2, 2025.

Final Output — Canonical FrameStore



Canonical FrameRecord

frame_id, video_id, authoritative frame_idx
keyframe_order, timestamp_ms, image/thumbnail paths
width, height, optional shot_id, is_anchor

Stable library boundary

```
get(frame_id)  
get_many(frame_ids)  
get_neighbors(...)  
filter_frame_ids(...)
```

Scope boundary: preprocessing stops after validated FrameStore creation.